

■ Closing & Master Data Modules & Business Cycles

D365FO Middleware Backend — Month-End Period Closing & System Master Data

Target Audience: Chief Accountants, Financial Controllers, Enterprise Architects.

1. Closing Module Business Overview

The **Closing** module manages period-end accounting entries and General Ledger (GL) reconciliations:

- **Freight Period Closing (GL-Freight):** Adjusting revenue/expense accruals and clearing ocean/air freight transit main accounts at month-end.
- **Fleet/Trucking Period Closing (GL-Fleet):** Reconciling fleet fuel, driver trip allowances, truck maintenance accruals, and hired truck owner settlements.
- **Freight Difference Adjustments:** Allocating exchange rate differences and tariff rounding variances.
- **Custody Settlement Entries:** Reconciling employee/driver custody advance clearing accounts against verified trip receipts.

2. Closing Entry Processors & Business Scope

Processor Class Name	EntryProcessorTypes	D365FO Target Journal	Real-World Business Purpose
ClosingFreightEntryProcessor	LedgerFreightClosingEntry	GL-Freight Journal	Month-End Freight Revenue/Expense GL Closing
ClosingTruckingEntryProcessor	LedgerTruckingClosingEntry	GL-Fleet Journal	Month-End Fleet & Trucking GL Closing
ClosingFreightDifferenceEntryProcessor	LedgerClosingFreightDifference	GL-Freight Journal	Period Freight Variance & FX Differences
ClosingCustodySettlementEntryProcessor	LedgerCustodySettlementEntry	Custody GL Journal	Driver & Employee Custody Clearing

3. Financial & Accounting Impact in D365FO

■ Period Closing Journal (Ledger JournalHeaders):

- **DEBIT / CREDIT:** Expense Accrual Main Accounts (e.g. **510901** Freight Accruals)
- **DEBIT / CREDIT:** Clearing Main Accounts (e.g. **110901** Transit Clearing)

- **Voucher Balance Requirement:** Total Debits MUST strictly equal Total Credits (**Sum(Debit) == Sum(Credit)**) for every voucher. Imbalanced vouchers are rejected before submission.

4. Master Data Module Overview

The **Master Data** module provides centralized caching and synchronization for D365FO enterprise entities:

Key Master Data Services:

1. **Financial Dimensions Service:** Validates and caches the 15 financial dimension tags (**MainAccount**, **BusinessUnit**, **CostCenters**, **Activity**, **Location**, **SalesMan**, **CoordinatorMan**, **Direction**, **Vendor**, **SubVendor**, **Worker**, **TruckerType**, **TruckNumber**).
2. **Main Accounts Cache:** Stores chart of accounts definitions, account types (**Ledger**, **Cust**, **Vend**, **Bank**), and posting restrictions.
3. **Exchange Rates Service:** Fetches monthly and daily exchange rates for **USD** and **EUR** to **EGP**, ensuring precise reporting currency conversions.
4. **Tax & WHT Service:** Caches Sales Tax Groups (**VAT_14**) and Withholding Tax rules (**WHT_1%**, **WHT_3%**).

5. End-to-End Closing Business Cycle Flow

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[Month-End Accounting Cut-off Reconciled by Finance]
■
▼
[API Endpoint: Upload Closing GL Excel]
■
▼
[ClosingFreightEntryProcessor]
(Verifies Voucher Balance: Sum(Debit) == Sum(Credit))
■
▼
[DataBatch Status: PENDING (Max 1000 Lines)]
■
▼
[BullMQ Queue: dfo-ledger-journal-queue]
■
▼
[LedgerJournalPostingStrategy]
■
▼
[Post OData to D365FO LedgerJournalHeaders]
■
▼
[D365FO Creates Posted General Ledger Journal]
(Posts to GL-Freight or GL-Fleet Journal)
■
▼
[Batch Status Set to COMPLETED]

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Detailed Step-by-Step Cycle Execution:

1. **Closing Sheet Preparation:** Accounting team prepares month-end journal entries in Excel.

2. **Strict Balance Audit:** The processor verifies that every voucher balances to zero (**Sum(Debit) - Sum(Credit) == 0**). If an imbalance exists, the upload fails immediately with line references.
3. **Queue Submission:** Dispatched to **dfo-ledger-journal-queue**.
4. **OData Execution:** Posts header and line records into D365FO **LedgerJournalHeaders** for journal names **GL-Freight** or **GL-Fleet**.
5. **Period Finalization:** D365FO posts the general ledger journal, completing month-end closing.